Did Americans Stop Moving Toward Opportunity?

Income, migration and the stalling of big cities, 1880–2020
(progress report)

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DATA320

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Abstract

For most of US history, people moved toward the places where incomes were highest. This project asks whether that is still true and, if not, when it changed. With state population and income data from 1880 to 2020, I estimate how much faster richer states grew than poorer ones, and I test whether that relationship was the same before 1970 and after 1990. It was not: before 1970 a state with 10% higher income grew about 1.7 percentage points faster per decade, but from 1990 on it grew slightly *slower*. The difference is statistically significant ($\chi^2(1) = 19.0$, $p < 0.001$). About 95% of the change comes from migration, not from births and deaths. A second analysis of 81 large metropolitan areas shows that the well-known stalling of big cities after 1950 was at first a move to the suburbs within growing regions. Only from the 1970s did whole metro areas fall behind, almost all of them in the Northeast and Midwest. The next stage of the project will test whether housing costs and land-use rules explain these patterns.

1 Questions

Housing has become expensive in the US cities with the strongest economies. One explanation is that rules limiting new housing keep people from moving to those cities. Before testing that explanation, the project first establishes the basic facts it has to explain. This report covers two of them:

Q1. Does population still grow faster in richer places, and when did that change? If housing limits are keeping people out of high-income places, the historical tendency of people to move toward higher incomes should have weakened.

Q2. When big cities stopped growing after 1950, did their whole region stop growing, or did people just move to the suburbs? City population counts alone can make a growing region look like it is in decline.

2 Data

All data are public. Table 1 lists the sources. The data pipeline is written in Python; it downloads or reads each source, converts it into one common long format, checks it and builds the analysis panels.

There are 50 states plus DC. A *metro* is a 2023 metropolitan area, built from its counties, so its boundary stays the same in every year. The city analysis uses the **81 metros that contained one of the 100 largest US cities in 1950**; "the city" is that metro's largest 1950 city.

Table 1: Data sources.

Data	Source	Coverage
State population	US Census via IPUMS NHGIS	1880–2020, every 10 years
State income per person	Klein (2013); Easterlin (1957); BEA	1880–1920; BEA 1929 on
State population and income, every year	Bureau of Economic Analysis (BEA)	1929–2024
County net migration	UW–Madison Applied Population Lab; ICPSR	1950s–2010s, by decade
City population	Stanford CESTA; IPUMS NHGIS	1900–2020
City land area	Census Working Paper 27 (Gibson, 1998); Census Gazetteer files	1950–2020
Metro areas (counties in each)	OMB 2023 metro definitions	held fixed for all years

3 Methods

3.1 How strongly does growth follow income?

For each period, I regress population growth on income:

$$\underbrace{\ln P_{i,t+1} - \ln P_{i,t}}_{\text{growth of state } i} \;=\; \alpha_t \;+\; \beta \cdot \underbrace{\ln(\text{income per person})_{i,t}}_{\text{how rich state } i \text{ is}} + \varepsilon_{i,t}. \tag{1}$$

The slope β measures how much faster richer states grow. All slopes in this report are scaled to growth *over ten years*. For example, $\beta = 0.17$ means that a state with 10% higher income grows about $0.17 \times \ln(1.1) \approx 1.7$ percentage points faster per decade. A positive β means people are moving toward income. A β near zero means growth no longer depends on income. The term α_t is a separate intercept for each year. It absorbs anything that hit the whole country at once, such as the baby boom, wars and recessions, so β compares states only with each other in the same year.

3.2 The statistical test

Model. I use the yearly BEA data (1929–2024) and let the slope differ across three periods:

$$g_{i,t} = \alpha_t + \beta_{\text{before}}\, x_{i,t}\, \mathbf{1}[t < 1970] + \beta_{\text{middle}}\, x_{i,t}\, \mathbf{1}[1970 \le t < 1990] + \beta_{\text{after}}\, x_{i,t}\, \mathbf{1}[t \ge 1990] + \varepsilon_{i,t}, \tag{2}$$

where $g_{i,t}$ is growth from year t to $t+1$ and $x_{i,t}$ is log income per person. The symbol $\mathbf{1}[\cdot]$ is the *indicator function*: it equals 1 when the condition in brackets is true and 0 when it is false,

$$\mathbf{1}[t < 1970] = \begin{cases} 1 & \text{if } t < 1970, \\ 0 & \text{otherwise.} \end{cases}$$

In any given year exactly one of the three indicators equals 1, so each year's growth is explained by one slope only. For example, in 1955 the equation reduces to $g_{i,1955} = \alpha_{1955} + \beta_{\text{before}}\, x_{i,1955} + \varepsilon_{i,1955}$. There are 4,803 state-years: 51 states (including DC), growth years 1929–2023. The 1970 and 1990 cut points come from earlier break-date searches in this project and from the literature (Ganong and Shoag, 2017).

Hypotheses.

$$H_0 : \beta_{\text{before}} = \beta_{\text{after}} \qquad \text{vs.} \qquad H_1 : \beta_{\text{before}} \neq \beta_{\text{after}}.$$

Test statistic. A Wald test:

$$W = \frac{(\hat{\beta}_{\text{before}} - \hat{\beta}_{\text{after}})^2}{\widehat{\text{Var}}(\hat{\beta}_{\text{before}} - \hat{\beta}_{\text{after}})} \sim \chi^2(1) \text{ under } H_0.$$

The variance is *clustered by state*. The same state appears up to 95 times, and its yearly errors are not independent: a state that grows quickly one year tends to grow quickly the next. Treating the 4,803 rows as independent would make the test too confident. Clustering treats each state's whole history as one independent unit, so there are effectively 51 observations, and it also allows the error variance to differ across states. The significance level is $\alpha = 0.05$.

3.3 Where the change comes from

Over a decade, a state's growth is the sum of net migration and natural increase (births minus deaths). Net migration is measured directly, by county. Natural increase is the remainder, which also picks up any change in how completely the census counted people. Because regression is linear in the outcome, β for total growth splits exactly into β for migration plus β for the remainder. That shows how much of the change in β comes from people moving.

3.4 Cities versus their metros

For each metro and each period (1950–70, 1970–90, 1990–2020), I ask whether the city and the whole metro *kept up*, meaning they grew by at least half as much as the US did over the same period. US growth was 34% in 1950–70, 22% in 1970–90 and 33% in 1990–2020. Each metro then falls into one of four classes:

	Metro kept up	**Metro fell behind**
City kept up	growing	city-led
City fell behind	suburbanizing	stagnant

Suburbanizing means the city stalled while its region kept growing: people moved to the suburbs. *Stagnant* means the whole region fell behind. A city is *annexing* if its land area grew more than 20% over the period, so new residents partly came from moving the city line; otherwise it has *fixed* boundaries. Ranges around each share come from the bootstrap: I resample the metros with replacement 999 times and report the middle 90% of the results. The decision rules (what share counts as "mostly suburbanization," and so on) were written down before any result was computed, so the results could not shape them.

4 Results

4.1 Growth stopped following income (Q1)

How to read Figure 1. The vertical axis shows how much faster a richer state grew. Points above the zero line mean that people moved toward higher incomes; points near zero mean that income no longer predicted growth. From 1880 to the 1960s, every decade's dot is above zero, between 0.03 and 0.28. In the 1970s the dot falls below zero, it rises again in the 1980s, and from

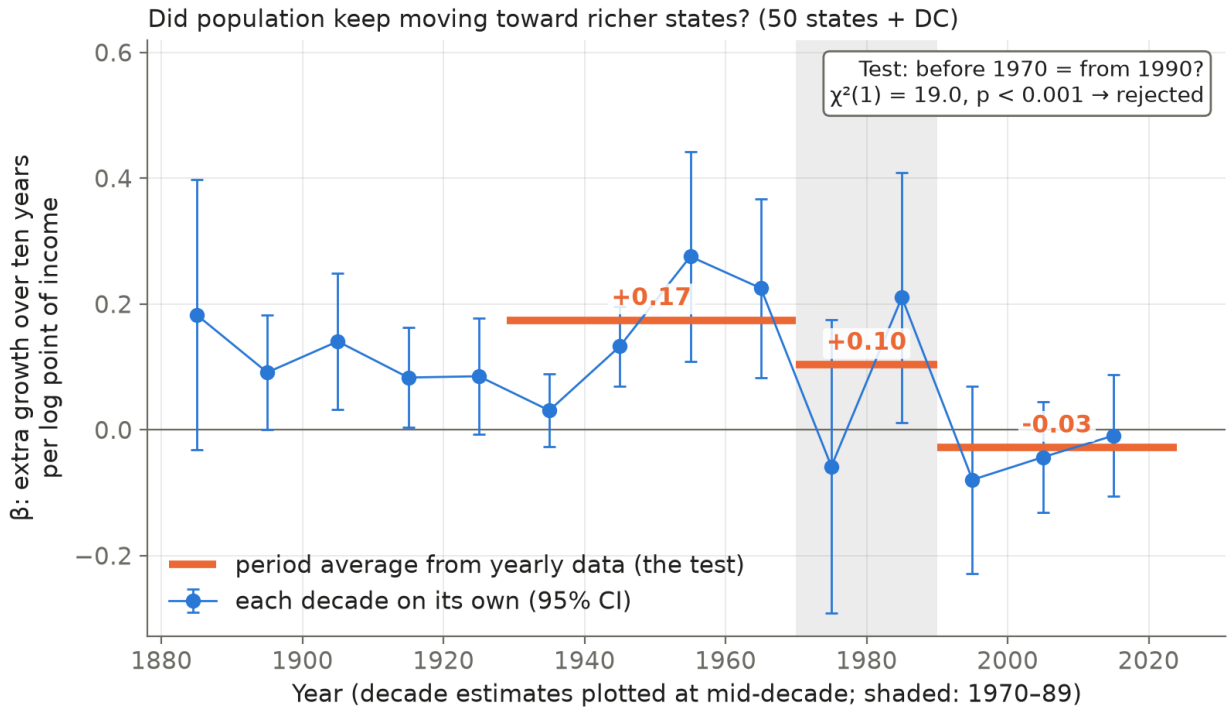


Figure 1: Population growth toward richer states, 1880–2020. Each **blue dot** is the slope β from equation (1), estimated on one decade alone (51 states, plotted at the middle of the decade), with a 95% confidence interval. The three **orange bars** are the period slopes from the test model, equation (2), using yearly data. The grey band marks 1970–89. The box reports the test of whether the first and last orange bars are equal.

the 1990s on every dot is at or below zero. The orange bars summarize this: +0.17 before 1970, +0.10 in 1970–89 and −0.03 from 1990. Some decades have wide intervals because each blue dot uses only 51 data points, which is why the test pools many years.

Test result. Table 2 gives the estimates. The slope fell by 0.20 (95% confidence interval 0.11 to 0.29), with $W = 19.0$ and $p = 1.3 \times 10^{-5}$. **We reject** H_0: the link between income and growth before 1970 differs from the link after 1990. In practical terms, a state with 10% higher income grew about 1.7 percentage points faster per decade before 1970, and about 0.3 points *slower* after 1990. This matches the pattern Ganong and Shoag (2017) found with different methods.

Is the result fragile? I re-ran the test six other ways. Each version changes one assumption that could plausibly drive the result. The test rejects H_0 at the 1% level in all of them (Table 3).

When did it change? The test establishes that a change happened, not exactly when. The middle period, 1970–89, is not significantly different from either side ($p = 0.42$ against "before" and $p = 0.16$ against "after"). The decade dots show why: the 1970s look like the period after, while in the 1980s people again moved toward income. Decade-level migration data put the main shift at about 1970. Yearly total growth cannot decide between 1970 and 1990.

The change comes from people moving. Table 4 splits the decade slopes into net migration and the rest. Of the 0.37 drop in the total slope between the 1950s–60s and the 1990s–2010s, 0.35,

Table 2: The test: income slope by period (yearly state data, 1929–2024; slopes on the ten-year scale; standard errors clustered by state).

$\hat{\beta}_{\text{before}}$ (1929–1969)	0.174
$\hat{\beta}_{\text{middle}}$ (1970–1989)	0.103
$\hat{\beta}_{\text{after}}$ (1990–2023)	−0.028
Difference, before − after	0.203
Standard error	0.046
95% confidence interval	0.112 to 0.294
Wald statistic W (χ^2, 1 d.f.)	19.0
p-value	1.3×10^{-5}
Observations (state-years) / clusters (states)	4,803 / 51

Table 3: The same test under other assumptions.

Version	p-value
Main (year effects only)	1.3×10^{-5}
Oil-producing states get their own year effects in 1973–1990 (oil boom and bust)	1.3×10^{-5}
. . . same, with a hand-picked list of oil states	1.2×10^{-5}
. . . same, oil years 1973–1986 only	1.3×10^{-5}
States compared only within their census division (9 regions)	0.008
Income standardized within each year (with oil-year effects)	4.9×10^{-8}
World War II years (1940–46) left out (with oil-year effects)	2.2×10^{-4}

or 95%, is migration. Births and deaths followed income only weakly in every period and barely changed.

Table 4: Income slope of decade growth and its two parts (50 states and DC; the slopes add up exactly: total = migration + the rest).

	1950s–60s	1970s–80s	1990s–2010s
Total growth	0.320	0.096	−0.047
Net migration	0.291	0.065	−0.058
Births − deaths (+ census coverage change)	0.029	0.031	0.012

4.2 Big cities: suburbs first, whole regions later (Q2)

How to read Figure 2. A value of 0.5 means half the metro lived in the main city. In 1940 the typical big city held about half its metro's population; by 2020 it held about a quarter. The fall is steepest between 1950 and 1980 and flattens after 2000. The share can fall either because the city lost people or because the suburbs grew faster, so this figure alone can't separate decline from suburbanization. Figure 3 does. Cities that annexed land, such as Houston and Austin, kept a higher share for longer.

How to read Figure 3. Read the colors from the bottom up. In 1950–70 (left panel), the largest group is orange, *suburbanizing*: 43 of 81 metros (53%) kept growing while their main city

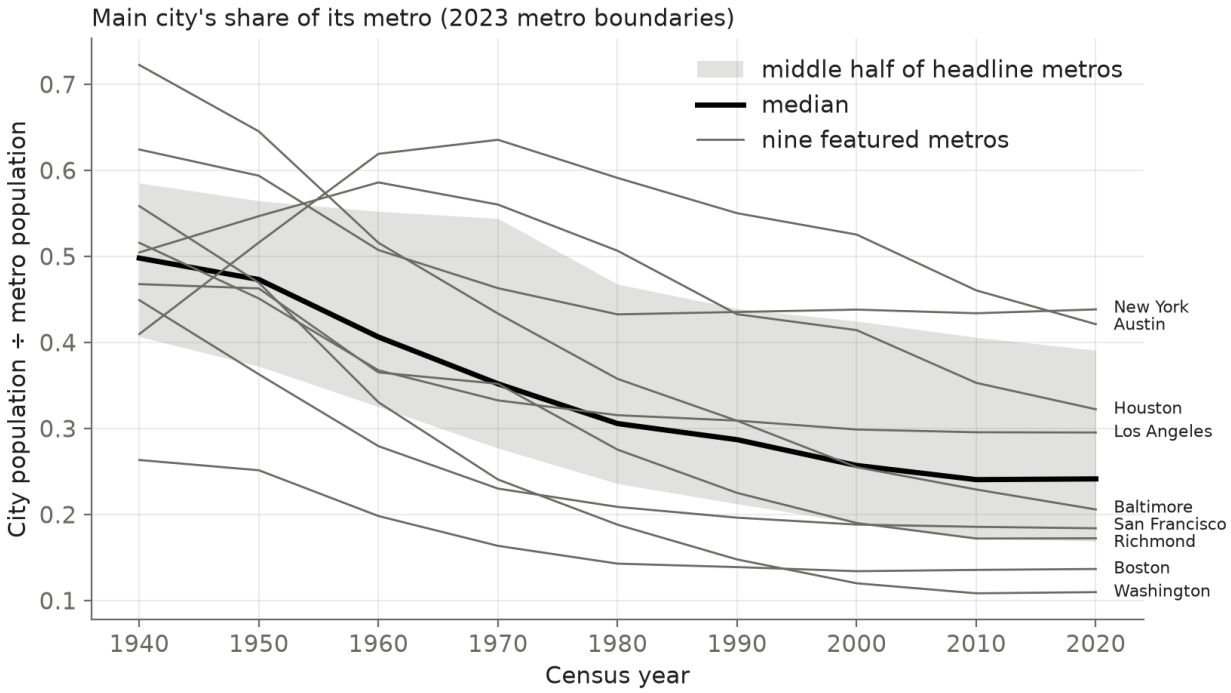


Figure 2: The main city's share of its metro's population, 1940–2020. Each thin grey line is one of nine featured metros, labelled at the right. The **black line** is the median across all 81 metros (the "headline" metros: those with a 1950 top-100 city), and the **grey band** holds the middle half of them (25th to 75th percentile). Metro boundaries are the 2023 definitions throughout, so a falling share means the city grew more slowly than the rest of its region, not that the region was redrawn.

fell behind. Green, *stagnant* (the whole metro fell behind), is only 6 metros. In 1970–90 green becomes the largest group, 35 of 81 (43%). In 1990–2020 it shrinks a little, to 28 of 80, while blue (*growing*, city and metro both kept up) recovers. The right panel, with a fixed 5%-per-decade bar, shows the same pattern, so it does not depend on how the bar is defined.

Results, with 90% bootstrap ranges.

- **Mostly suburbanization in 1950–70.** Of the 49 cities that fell behind, 88% (79–96%) were in metros that kept up. It is 78–93% under every alternative definition.
- **Fixed-boundary cities fell behind; annexing cities did not.** Among metros that kept up in 1950–70, 91% of fixed-boundary cities fell behind, against 22% of annexing cities. The difference is 0.69 (0.54–0.82). Part of this is built in, since people in annexed land count as city growth.
- **Whole metros fell behind later.** The share of metros that fell behind rose from 9% (1950–70) to 43% (1970–90), then 36% (1990–2020). The rise is 0.35 (0.25–0.44). It is almost all in the Northeast and Midwest (12% → 77% → 61%); in the South and West it stayed at 5–8%.

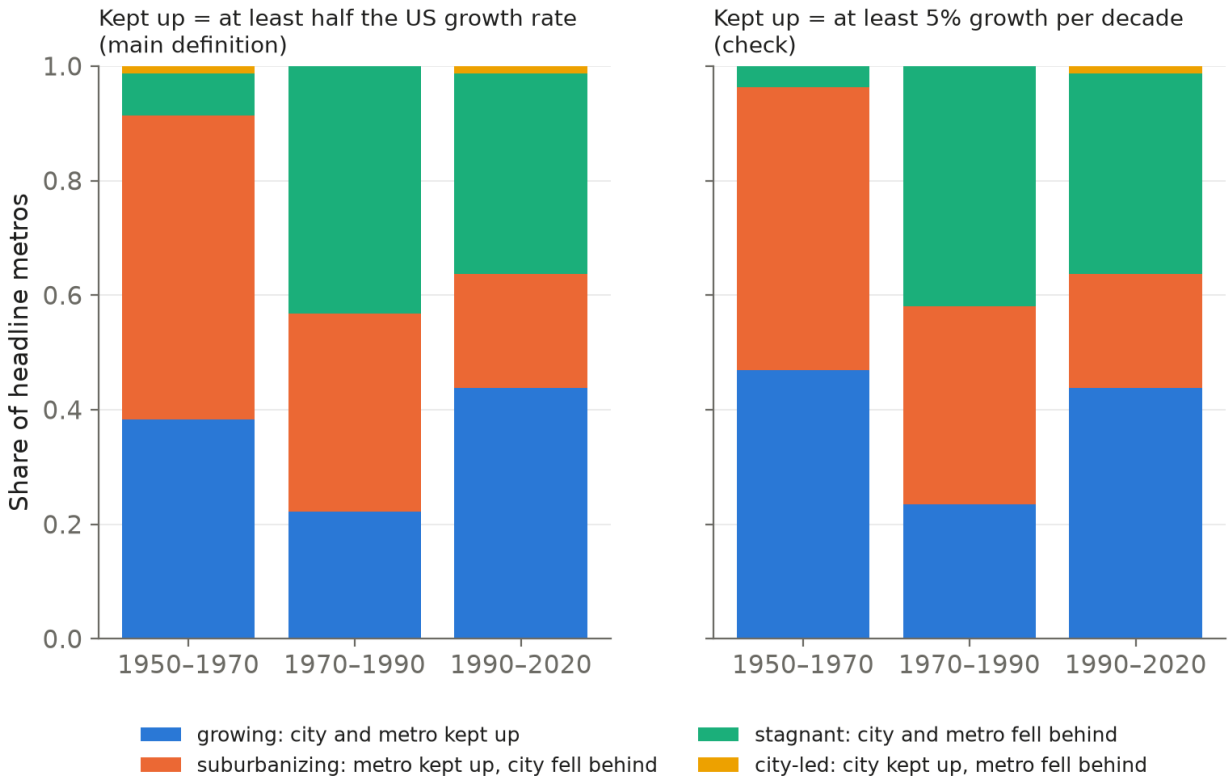


Figure 3: How the 81 big-city ("headline") metros grew, by period. Each bar divides the metros into the four classes defined in Section 3.4; the legend repeats the definitions. Bar heights are shares, so each bar sums to 1. **Left:** "kept up" means growth of at least half the US rate over the period (the main definition). **Right:** "kept up" means at least 5% per decade, a fixed bar that does not depend on how fast the US grew (a check).

5 Limitations

- **Association, not cause.** The test shows that the link between income and growth changed. It does not show *why*. Housing costs are one explanation; others include retirement migration to the Sun Belt, air conditioning, and the decline of manufacturing, which the regional pattern in Section 4.2 suggests: the Northeast and Midwest were hit hardest.
- **Timing.** The data agree that the change happened, but not on whether 1970 or 1990 is the better date.
- **Income data before 1929** come from historical estimates (Klein; Easterlin) that are less precise than BEA's; the test itself uses only BEA data from 1929.
- **Fixed 2023 metro boundaries** include counties that were rural in 1950, which can overstate early "suburban" growth.
- **Land areas** come from two sources that measure slightly differently (1950–80 and 1990–2020), which adds some noise to the annexing classification.

6 Next steps

1. Test the housing explanation directly: do metros that permit less housing (Census Building Permits Survey) or have stricter land-use rules (Wharton index) attract fewer people for the same income growth, and have their prices risen faster relative to income?
2. Compare with other countries (OECD metro data), to see whether the US is unusual.
3. A case study of a policy change: Auckland's 2016 upzoning and the construction that followed.

Use of AI tools

This report was written with the help of Claude Code, an AI coding assistant made by Anthropic. It was used to write and test the data pipeline and analysis code, to make the figures and to draft the text of this report.

Reproducibility

The project is a Python package with a data pipeline, two analysis notebooks (`1.1-lr-us-structural-break` for Q1, `1.0-lr-city-vs-metro` for Q2) and 248 automated tests that check both the code and the data, for example that city populations match published Census figures. Every number in this report comes from tables that the notebooks write to `reports/tables/`.

References

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